T-Tests and Chi-Square Tests

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1. What do these tests actually mean?

A t-test and a chi-square test are statistical methods used to study sample data. They help us decide whether a pattern in the sample provides evidence of a real difference or relationship in the larger population, rather than simply reflecting random variation.

Easy way to remember: A t-test compares numerical averages; a chi-square test works with category counts.

2. What is a t-test?

A t-test checks whether a difference between averages is statistically significant. A common example compares the mean exam marks of two independent groups.

Study method	Average exam marks
Method A	80
Method B	70

The sample averages differ by 10 marks. A t-test helps assess whether that observed difference is larger than we would expect from ordinary sampling variability. It also considers the variation in individual marks and the number of students in each group.

Important: A statistically significant result does not prove that Method A caused higher marks. For a causal claim, we must also consider how the groups were selected and other possible factors.

3. What is a chi-square test?

A chi-square test studies counts in categories. One type, called a chi-square goodness-of-fit test, checks whether the observed counts differ from an expected distribution.

Example: 100 students choose their favorite programming language. If all four languages are equally popular, we would expect 25 students in each category.

Language	Observed students	Expected if equally popular
Python	40	25
Java	30	25
C++	20	25
JavaScript	10	25

The observed numbers are not equal. The chi-square test helps decide whether the differences provide enough evidence to conclude that the languages are not equally popular in the population.

4. What is a p-value?

A p-value tells us how surprising a result at least as extreme as the one observed would be if the null hypothesis were true. The null hypothesis is usually a statement of no difference or no relationship.

If $p < 0.05$: reject the null hypothesis at the 5% significance level; the result is statistically significant.

If $p >= 0.05$: do not reject the null hypothesis; there is insufficient evidence at that level.

A p-value is NOT the probability that the null hypothesis is true. A result that is not significant also does not prove that the groups are identical.

5. Key difference

T-test	**Chi-square test**
Compares means of numerical data	Analyzes counts of categorical data
Example: average salary or exam marks	Example: preferred language or payment method
Produces a t-statistic and a p-value	Produces a chi-square statistic and a p-value

6. Why does a data analyst use them?

T-test: Compare average customer spending between customers offered a discount and those not offered one.

Chi-square test of independence: Investigate whether payment-method preference is associated with age group.

These tests support evidence-based decisions, but results should also be checked for practical importance, appropriate data collection, and test assumptions.

Final takeaway: T-test = compare averages. Chi-square test = analyze category counts. In both cases, use the p-value and context to explain the result.